CMP6200/DIG6200
Individual Undergraduate Project (FYP)
Interim Progress Review
AR based Virtual Tour Guide

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Course: CMP6200 Individual Honours Project

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Reporting Period: February 1,2026 to April 24,2026

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1. Log of Completed Activities & Milestones

This report is the compilation of all the technical and non technical work that I have done during the period of the review of February 1,2026 to April 24,2026

Aspect	Description
Current Project Phase	Advanced development and early integration phase, transitioning from experimentation to a working prototype
UI Development	Core user interface fully completed and functional
Chatbot (NLP Integration)	Functional chatbot developed using API-based NLP, enabling interactive user queries
Computer Vision Model	Landmark recognition model successfully trained using prepared dataset
AR Development (Unity)	Partially implemented; currently testing camera feed and real-time AR overlay behavior
System Integration Status	In progress; focus on connecting CV, AR, and chatbot into a unified system
Development Approach	Shift from isolated module development to integrated system architecture
Key Achievement	Successful development of all core components required for the system
Current Focus	Ensuring seamless interaction between modules and system-level functionality
Significant Insight	Identified and addressing the challenge of integrating multiple AI technologies into one cohesive application
Alignment with Proposal	Directly addresses the problem of fragmented AI systems in tourism applications

Figure 1: Project Progress Overview

2.Artefact Design and Development

This section shows the progress made so far and highlights the challenges and how they were resolved during the project lifecycle.

2.1 Progress Against Original Timeline

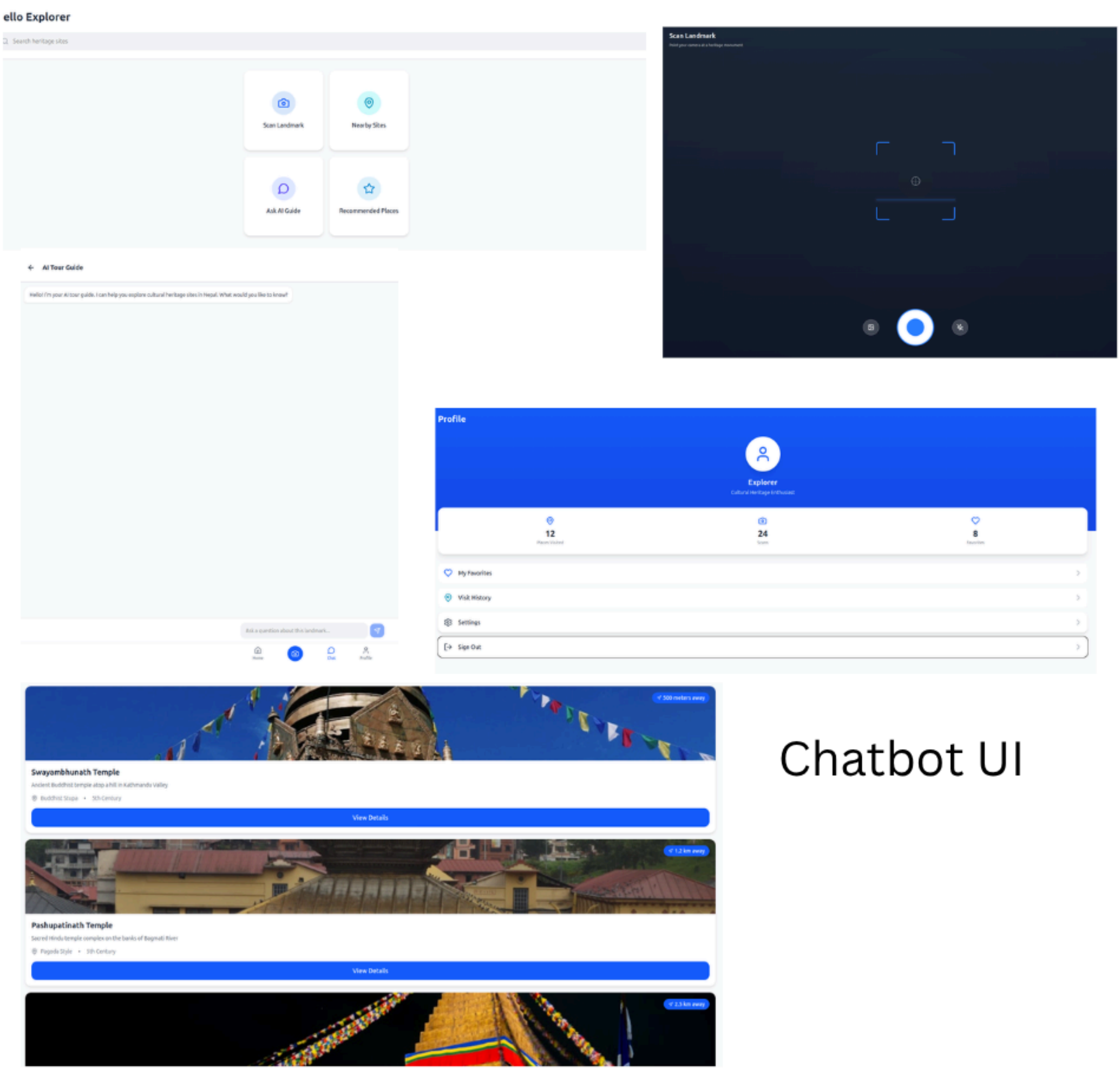
Component	**Status**	**Description**
Dataset Collection	Completed	3500–4000 images across all heritage sites
Image Preprocessing	Completed	Augmentation, resizing, normalization
Model Training (CV)	Completed	Lightweight model trained for mobile deployment
UI Design	Completed	Fully functional mobile interface
Chatbot (NLP API)	Completed	Context-aware chatbot integrated
AR Environment (Unity)	In Progress	Camera + AR overlay under testing
System Integration	In Progress	Components developed separately, integration pending
Testing & Validation	In Progress	Initial testing done, full testing pending

Figure 2: Progress Against Original Timeline

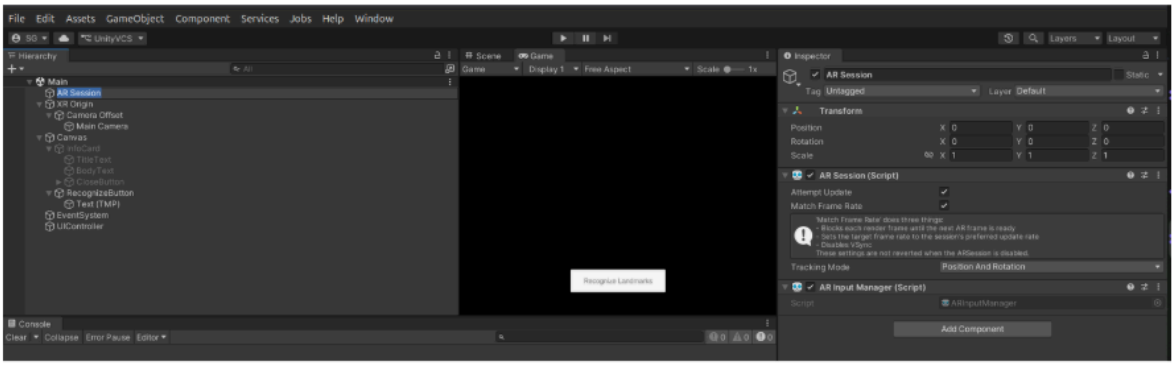
Based on the Gantt chart and completed milestones, the project is currently:

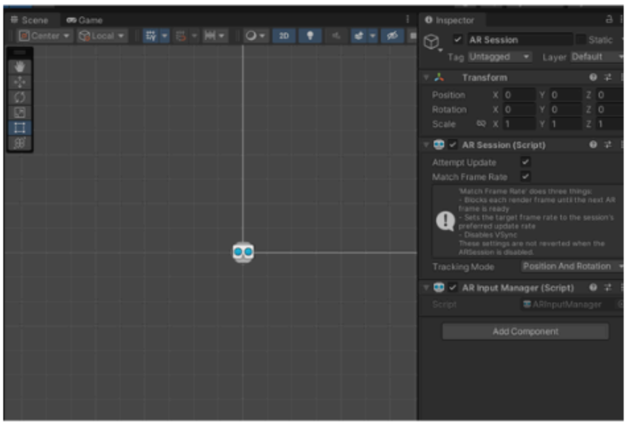
☑ On Track

2.2 Implementation and Current State



Chatbot UI





Unity Interface and Scene play

2.4 Challenges Encountered & Resolutions

While doing this, I faced many challenges, mainly related to technical issues and resource constraints. These challenges and resolved actions are summarised below:

Challenge	Impact	Resolution
Fragmented development	Delayed integration	Shifted to integration-focused approach
Unity AR complexity	Slower development	Incremental testing in Unity scenes
Multi-tech stack (CV + AR + NLP)	High system complexity	Modular debugging strategy
Time pressure	Risk to completion	Prioritizing core features only

Figure 4: Challenges Encountered & Resolutions

3. Action Plan: Goals for Next Period

- I will complete full system integration by connecting the computer vision model, AR module (Unity), and chatbot into a single working application.
- I will finalise the Unity AR environment, ensuring stable camera tracking and accurate real-time overlay of information.
- I will conduct system testing, including functional testing and validation of real-time performance.
- I will improve and evaluate the landmark recognition model using unseen test data to ensure reliability.
- I will implement a basic content-based recommendation system aligned with the project objectives.
- I will complete report writing, focusing on implementation, results, and evaluation sections.
- I will prepare a working prototype/demo for the final presentation and viva.

4. Questions for Supervisor

1) Which evaluation metrics would you recommend for assessing overall system performance, particularly combining model accuracy, AR responsiveness, and user experience?
2) Is achieving approximately 80% accuracy in landmark recognition sufficient for this project, or should I focus more on improving robustness under real-world conditions

5. Conclusion and Reflection

The project has already moved from an idea through the design and preparation phase, and into the core technical phase of the project with most parts designed. The present emphasis on integration is a vital and likely phase of the system development process, where planning is confronted with execution.

The project has shown that developing a smart AR system is not about technologies alone, but how they are integrated to provide a seamless user experience. This is in line with the project's goal of resolving isolated AI systems in tourism.

The project has a plan, priorities and is well on the way to successful completion within the timeboxed project duration, as long as integration and testing goes well.